

Introduction to Machine Learning

Course Name	Course type (credit/hours)	Required course(3/3)		
	Course code	F127	Subject code	AAI332
	Target students Division/major/grade	Applied Artificial Intelligence/	Opening semester	2026 1ST SEMESTER
	Class time and classroom	Tue E(IUC207)Fri E(IUC207)	English Grade	
Reference to this course	Prerequisite courses	자료구조		
	Related basic courses	빅데이터개론및분석		
	Recommended concurrent courses			
	Related advanced courses			

Instructor	Name (division)		Han Seung-Hyun(Applied Artificial Intelligence)			
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Teaching Assistant	Name (title/division)					
	Office Room Number		Office phone Number		e-mail	

1. Introduction

This class offers a theoretical foundation in machine learning's essential concepts and algorithms. We examine the architectures and distinctions between supervised and unsupervised learning, covering key areas like regression, classification, clustering, and dimensionality reduction. Students investigate real-world algorithm applications via case studies and visual aids. The format features lectures, quizzes, assignments, group discussions, and project presentations.

2. 교육목표와 교과목 학습성과

순번	교육목표와 성과관리	하위역량1	하위역량2	하위역량3
1	Theoretically understand basic concepts and algorithms in machine learning.			
2	Grasp the structures and differences between supervised and unsupervised learning.			
3	Explain principles and characteristics of major algorithms (regression, classification, clustering)			

3. 교과목과 핵심역량 간 연계

대학 핵심역량	감수성	소통력	창의성	사고력	개방성
	☐	☒	☐	☒	☒

4. Class types and activities

This course centers on theoretical understanding of foundational concepts and key algorithms in machine learning, supplemented by hands-on practice using ML tools and real-world case studies. Students aim to grasp the overall structure of machine learning and develop the ability to select and apply appropriate models.

Course Learning Outcomes

Students who successfully complete this course will achieve the following learning outcomes:

- Distinguish and explain the definition of machine learning and various learning types (supervised learning, unsupervised learning, etc.).
- Understand the operating principles and characteristics of major algorithms such as regression, classification, and clustering.
- Select suitable machine learning algorithms for real-world cases and propose application strategies.
- Interpret basic performance metrics for models (accuracy, precision, recall, etc.).
- Develop skills to apply theory, summarize, and present through simple team projects.

Class Format and Delivery Method

Classes are primarily conducted in lecture format, with theoretical explanations demonstrated using machine learning tools (e.g., scikit-learn, PyTorch).

Students gain practical experience by implementing and experimenting with models through homework and projects. Project results are summarized in presentations, accompanied by group discussions to enhance knowledge sharing and critical thinking.

4.1 수업평가 문항선택

- | | | |
|----------------------------------|------------------------------|------------------------------------|
| ☐ 일반(기본) | ☐ 강의식 | ☐ 대형강의(AFL) |
| ☐ 학습자활동중심 | ☐ 실험 | ☐ 사이버강의 |
| ☐ 플립드러닝 | | |

5. Support Systems in Use

- | | |
|--|---|
| ☒ lecture | ☐ 문제 풀이 |
| ☐ 글쓰기 | ☐ discussion and debate |
| ☐ 발표 | ☒ team project(presentation and case studies) |
| ☒ experiments(role-playing, etc) | ☐ designing and production |
| ☐ on-site learning(on-site training) | ☐ 사전 학습 |
| ☐ others | |

6. 수업방식/수업방법

- ☒ 대면수업
- ☐ 비대면수업
- ☐ 혼용수업(대면+비대면)
- 수업방법
- ☒ 이론
- ☐ 실험+실습
- ☐ 동영상강의100%
- ☐ 실시간화상강의
- ☐ 동영상+실시간
- ☐ 대면+동영상
- ☐ 대면+동영상+실시간
- ☐ 대면+실시간
- ☐ 대면+실시간(강의실 스트리밍)

7. Teaching Tools

- | | | |
|---|--|---|
| ☒ PBL(Problem Based Learning) | ☐ CBL(Case Based Learning) | ☐ TBL(Team Based Learning) |
| ☐ UR(Undergraduate Research) | ☒ FL(Flipped Learning) | ☐ DSAL(Data Science Active Learning) |
| ☐ others | | |

8. Knowledge and ability required for taking this course

기초적인 수학 및 소프트웨어 지식
기초적인 프로그래밍 언어 구현 능력 (예, 파이썬)

9. Method of Evaluation

Evaluation Item	The Number of Times	Evaluation Proportion	Remarks
Attendance		10	출석 및 수업 참여 태도. 학칙 근거 하에 출석 미달시 F
midterm exam		30	
final exam		30	
quiz			
presentation		10	프로젝트 최종 발표
discussion			
homework		20	프로젝트 제안, 수행 및 결과 보고서 및 성취도 평가
etc			
report			
study hours			

10. Textbook and supplementary material

Main/Sub	Title (Web-site)	Writer	Publisher	Publication year
Main	Data Science for Business	Foster Provost & Tom Fawcett	O'Reilly	2013
Sub	Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow	Aurelien Geron	O'Reilly	2019
Sub	Introduction to Machine Learning with Python	Andreas C. Muller & Sarah Guido	O'Reilly	2017
Ref.	Data Science from Scratch	Joel Grus	O'Reilly	2015
Sub	혼자 공부하는 머신러닝+딥러닝	박해선	한빛미디어	2022

11. Class system and Class shedule

빅데이터 및 인공지능 시대에 필수적인 데이터 사이언스와 머신러닝의 기초 지식부터 기술적 개념까지 체계적으로 학습한다.
다양한 비즈니스 문제를 기반으로 머신러닝 기술을 응용하는 문제 해결 방법론(methodology), 기술적 방법(method), 관련 이슈 및 개념을 익힌다.

지도 학습 (Supervised Learning)

Decision Tree: 구조, 분할 기준(예: Gini impurity, Entropy).

Feature selection 방법 및 Overfitting 방지 기법(Pruning, Regularization).

평가 지표: Accuracy, Precision, Recall, F1-score.

비지도 학습 (Unsupervised Learning)

Clustering: K-means, Hierarchical clustering; Similarity 측정(Distance metrics).

Nearest Neighbor 접근법 및 차원 축소(PCA).

실습 및 적용

머신러닝 툴(Scikit-learn 등)을 활용한 이론 구현 연습.

문제 정의 → 데이터 전처리 → 모델 실험 → 분석 과정 학습.

과제를 통해 비즈니스 문제에 적합한 툴 선택 및 실험 능력 배양.

< Class Schedule >

* language : K-korean, E-English

Weeks	Topics	language	Instructor	Teaching Method	Evaluation Method	Matter to be prepared
1	Intro to Data Science and Machine Learning	K	Han Seung-Hyun			
2	Experimental Settings	K	Han Seung-Hyun			
3	Data Management	K	Han Seung-Hyun			
4	Regression I – KNN, Linear Regression	K	Han Seung-Hyun			
5	Regression II – Multiple Linear Regression	K	Han Seung-Hyun			
6	Classification I – Logistic Regression	K	Han Seung-Hyun			
7	Classification II – Stochastic Gradient Descent	K	Han Seung-Hyun			
8	중간고사	K	Han Seung-Hyun			
9	Trees I	K	Han Seung-Hyun			
10	Trees II – Ensemble	K	Han Seung-Hyun			
11	Unsupervised Learning I	K	Han Seung-Hyun			

< Class Schedule >

* language : K-korean, E-English

Weeks	Topics	language	Instructor	Teaching Method	Evaluation Method	Matter to be prepared
12	Unsupervised Learning II	K	Han Seung-Hyun			
13	Deep Learning I	K	Han Seung-Hyun			
14	Deep Learning II	K	Han Seung-Hyun			
15	과제발표	K	Han Seung-Hyun			
16	기말고사	K	Han Seung-Hyun			

12. Other items of notification

동영상 강의는 주/보조 강의 자료로써 쓰일 예정임.

기타 강의에 대한 안내 및 공지는 AjouBB를 통해서 할 예정이나, 궁금한 사항이나 질문은 교수나 조교 이메일을 통해 연락 바랍니다.

13. Generative AI Usage Policy

※This generative AI usage policy is provided as a general guideline.

Depending on the specific educational objectives, content, and assessment methods of each course, instructors may, at their discretion, adopt these policies or establish separate protocols. .

Students are responsible for reviewing the course syllabus and adhering to the instructor's specific directions.

Applica tion crite ria	General Principles	Description	Student Responsibilities	Consequences of Violation
☐	AI-Free Assessment	The use of AI is prohibited in all coursework, including class activities, assignments, and exams	– Absolutely no AI usage – Learning objectives must be achieved independently	– Considered Academic Misconduct – Subject to strict disciplinary action according to school regulations, including assignment invalidation grade deductions, and academic warnings
☒	AI-Assisted Assessment	Permitted with instructor approval or with proper citation and usage disclosure.	– Request prior approval or document generative AI usage. – Documents must include prompts, date of use, tool names, and generated outputs	– Failure to cite sources or exceeding the approved scope of use constitutes academic misconduct
☐	AI-Integrated Assessment	Unrestricted use permitted, but requires critical analysis and creative contributions.	– Submission of raw AI output is prohibited – Must compare, verify, and refine AI results, adding personal interpretation – Transparently disclose AI usage, including prompt intent and the revision process	– Undisclosed use or submission of raw AI output is considered academic misconduct.

13.1 Guidelines for Generative AI Usage Policy

※Please specifically state the 'Generative AI Usage Principles' for this course to ensure the achievement of learning objectives and adherence to academic integrity.

– Permission Type (AI-Free / AI-Assisted / AI-Integrated Assessment), Specific Scope (Class activities, Assignments, Exams, Labs), Requirements for Use (Citation, Prompt submission), and Consequences of Violation (Grade deduction, Considered academic misconduct), etc.

실습 시 조건부 허용
출처 표기 및 프롬프트 제출
위반 시 부정행위 간주